

Cleaning Pipeline Report

1. Load and Align Data

- Loaded **2160 EEG trial files**.
- Original `labels.csv` had **2400 labels**.
- The last **240 extra labels were removed** after instructor confirmation.
- Final aligned dataset: **2160 trials**.

2. Sampling Rate

- Sampling rate was confirmed by the instructor as **250 Hz**.
- Each trial has **2500 samples**, which is about **10 seconds** of EEG data.

3. Time Column

- The original `Time` values were irregular and started from large absolute numbers.
- For processed data, time is rebuilt using:

0.000, 0.004, 0.008, ...

because:

$1 / 250 \text{ Hz} = 0.004 \text{ seconds per sample}$

4. Band-Pass Filtering

- Applied a **0.5–40 Hz Butterworth band-pass filter**.
- Goal:
 - remove very slow drift,
 - remove high-frequency noise,
 - keep useful EEG activity.
- The important motor-imagery bands are still preserved:
 - Mu: **8–12 Hz**
 - Beta: **13–30 Hz**

5. Notch Filter

- A separate 50 Hz notch filter was not considered necessary for the current 0.5–40 Hz pipeline because 50 Hz is already outside the kept frequency range.

6. Artifact Rejection

- EDA identified suspicious trials using:
 - unusually large peak-to-peak changes,
 - unusually large sudden jumps.
- Trials flagged by both checks were saved as rejection candidates.
- Rejected trials are excluded using the clean manifest.
- Their files are not permanently deleted.

7. Clean Dataset

- `clean_manifest.csv` contains the filtered trials accepted after artifact rejection.
- This is one valid dataset for modeling.

8. Common Average Reference (CAR)

- CAR was also tested as a separate preprocessing branch.
- At every sample, the average of:

FZ, C3, CZ, C4

is calculated and subtracted from each channel.

- Goal: reduce signal components shared by all channels and highlight differences between channels.
- Because only four EEG channels are available, CAR is treated as an experimental option rather than automatically better.

9. Re-Referenced Dataset

- `rereferenced_manifest.csv` contains the accepted trials after CAR.
- Modeling can therefore compare:

Clean filtered EEG

vs

Clean filtered EEG + CAR

10. PSD

- PSD was used during EDA to study frequency content and LEFT/RIGHT differences.
- PSD is not a mandatory permanent cleaning step.
- It may later be used for feature extraction such as Mu/Beta band power.

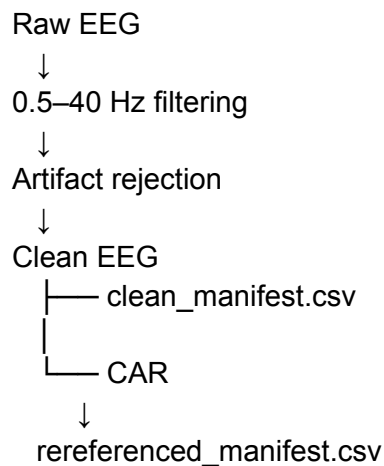
11. Normalization

- Global normalization was not applied permanently.
- Normalization should be fitted on the **training data only** after train/validation/test splitting to avoid data leakage.

12. Epoching and Baseline Correction

- These are not finalized yet.
- We still need the exact experimental timing that tells us:
 - when the LEFT/RIGHT cue appears,
 - when motor imagery starts,
 - when it ends,
 - whether a real baseline/rest interval exists.

Final Available Modeling Branches



Both branches should be compared during modeling using the same data split and evaluation method.